

Xinlin Zhang

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Career Objective

AI/Software Engineer.

Seasoned software engineer with a Master's Degree of AI in AUT. Passionate about real-world AI applications. Seeking to join a team in which I can apply my expertise in building innovative and cutting-the-edge AI systems.

Skills Summary

- proficient in Python, familiar with PyTorch
- **Computer Vision (CV):** Experienced in designing, implementing, training and evaluating various deep learning models from scratch by using PyTorch, specialized in building customized CNN architectures. Familiar with the techniques of R-CNN family and YOLO series models for object detection. Be able to reproduce the latest YOLO26 architecture which is exactly the same as the design of Ultralytics.
- **MLOps & Automation:** Capable of automating Label Studio into working flow, integrating SAM2 or self-trained model into Label Studio, realizing interactive model-assistant data labeling or auto-labeling with self-trained models.
- **Large Language Model (LLM):** Hands-on experience in deploying self-hosted LLMs and deep understanding of prompt engineering. Be able to work together with Claude Code.
- **Strong Computer Science Background:** expertise on Computer Science, OS, Network, Algorithm and Database.
- **Seasoned Software Engineer:** many years' experience in C/C++, familiar with Linux and Shell.
- [Personal Projects Portfolio](https://github.com/xinlin-z/portfolio) (github.com/xinlin-z/portfolio)

Professional Experience

AI Software Research Engineer Intern, TOMRA Food (New Zealand), 11/2025 – Present

- Working in the Data Scientist Team as an Intern.
- Design, Implement, Train and Evaluate highly modernized, customized and configurable deep learning models for multi-class image segmentation tasks. These models have fixed input and output, and need to be run in a memory-constrained real-time environment. The model's performance is good enough to test on the customer's site. (Python, PyTorch)
- Design and implement wrapper model to include post-processing computations and realized end-to-end pipeline. (PyTorch)
- Automate Label Studio into working flow, realize interactive model-assistant data labelling, and build training and validation dataset. (Python, Shell, Label Studio)
- Implement the whole working codebase, including models, data augmentation algorithms, training and fine-tuning scripts, and all other tools. (Python, PyTorch, Shell, Javascript, FastAPI)
- Solve/bypass a few critical issues related to GPU VRAM and onnx runtime (TensorRT).

Senior System Software Engineer, Astute-Tec (China), 2021 – 2023

- Designed and implemented a highly optimized and configurable BGRA image comparison algorithm and a hybrid H.264/JPEG transportation protocol, reducing CPU utilization from over 20% to below 5% in a cloud desktop system during static or minimal screen activity. (C++, Python)
- Designed and implemented a priority-based multi-TCP schedule algorithm, significantly improving user responsiveness and fluidity in a highly loaded cloud virtual desktop environment. (C++)
- Recipient of the “Network and Video Codec Expert Award”, May 2022.

Software Engineer, VP, Co-founder, GWorld (China), 2014 – 2021

- Solely designed and implemented an auto-production system for manufacturing company proprietary optical modules ($\leq 10G$) and OEM modules. (Python, SQLite)

Software Engineer, ZTE R&D Center (China), 2006 – 2013

- Developed and maintained many features in the TCP/IP stack (such as Ethernet, VLAN, Routing Protocol, ACL, and RTP) which were integrated into many different telecommunication products. (C)
- Recipient of the Department's “Best Programmer Award”, December 2009.

Education

Auckland University of Technology (New Zealand), Feb. 2025 – Jul. 2026

Master of Computer and Information Science. Graduated with Honours (First Class)

Coursera.org:

AI Agent Developer (Vanderbilt University), [Certificate](#) earned in Jun. 2025

Mathematics for Machine Learning and Data Science (DeepLearning.AI), [Certificate](#) earned in Feb. 2025

Northeastern University (China), Sep. 1998 – Jul. 2002

Bachelor of Engineering Degree in Management Information Systems (under the department of Computer Science)

Interests

Reading, Coding, Hiking and Green Tea.

Reference

Patrick Lindegger

Data Scientist in TOMRA Food

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Yanbin Liu

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